

# Simulation Based Inference

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Introduction

Methods

Pipeline

Evaluation

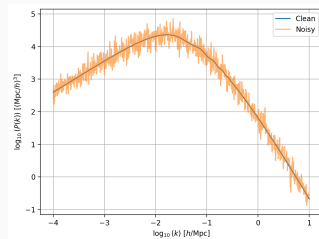
Summary

# Introduction

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## Task 6: Estimating Cosmological Parameters from Galaxy Power Spectrum

- The large-scale structure of the universe encodes vital information about:
  - The matter-energy composition
  - Expansion history
  - Initial conditions
- **Matter power spectrum,  $P(k)$**  : It describes variance in matter density as a function of the spatial frequency,  $k$ .



<https://wallpapercave.com/w/wp8201987>

The shape and amplitude of  $P(k)$  depend on following parameters:

- Hubble constant  $H_0$ ,
- matter density  $\Omega_m$ , and
- spectral index  $n_s$ .

**Objective:** infer the posterior distribution over cosmological parameters given noisy observations of the matter power spectrum  $P(k)$

# Methods

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# Initial information

## Initial suggestions:

1. Prior should reflect physically plausible ranges for each parameter (Planck, 2018):
  - $H_0 = 67.4 \pm 0.5$
  - $\Omega_m = 0.315 \pm 0.007$
  - $n_s = 0.965 \pm 0.004$
2. Noisy observations of matter power spectrum: generated using CAMB with added noise. (<https://camb.readthedocs.io/en/latest/CAMBdemo.html>)

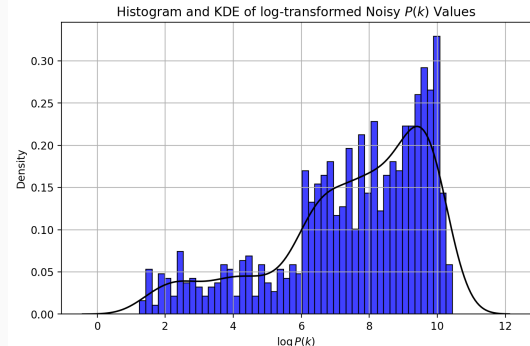
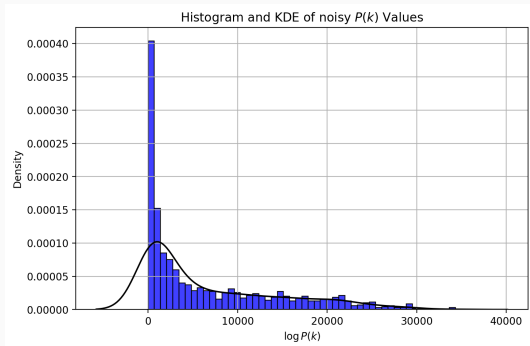
## Our assumptions:

1. Use uniform prior for each single parameter in  $\theta = \{H_0, \Omega_m, n_s\}$ ;
2. Parameters are distributed independently:

$$\theta \sim p(H_0, \Omega_m, n_s) = p(H_0) \cdot p(\Omega_m) \cdot p(n_s)$$

3. Noisy observations of  $P(k)$ : added Gaussian noise proportional to the spectrum values;

# Observed data



Right Skewed  $\Rightarrow$  We may try using log-transformed  $P(k)$  values for training.  
We will use  $\log_{10}(\cdot)$



# Pipeline

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1. Simulator
2. Adapter
3. Summary network
4. Inference network

Number of Simulations,  $S = 10,000$

Simulator:  $(\theta^{(s)}, y^{(s)}) \sim P(\theta, y), \quad s = 1, \dots, S$

```
simulator = bf.make_simulator ([prior, likelihood])
```

- Prior:  $H_0 \sim U[30, 100]$ ,  $\Omega_m \sim U[0.1, 0.6]$ ,  $n_s \sim U[0.8, 1.5]$
- Likelihood: CAMB algorithm using given parameter values of  $\theta = \{H_0, \Omega_m, n_s\}$  as well as  $k_{vals} = [10^{-4}, 10]$ ,  $\Omega_b = 0.05$  (baryon density)
- Likelihood for log-transformed  $P(k)$  values: same but  $\log P(k)_{\text{noisy}} = \log P(k) + \epsilon$ , where  $\epsilon \sim \mathcal{N}(0, 0.2)$

# Adapter and summary network

Adapter: to adapt the inputs and outputs of different steps of the process

```
adapter_std = (  
    bf.Adapter()  
    .concatenate(["H0", "Omega_m", "n_s"], into=  
"inference_variables")  
    .rename("log_P_k_noisy", "summary_variables")  
    .standardize("summary_variables"))
```

Summary network:  $q(\theta y) \rightarrow q(\theta h(y))$

Used architectures: DeepSet and MLP

```
summary_network_Deep = bf.networks.DeepSet()
```

First attempt: affine coupling in case posterior is simple

```
inference_network_affine = bf.networks.CouplingFlow()
```

Second attempt: spline coupling in case posterior is more complex

```
inference_network_spline =  
bf.networks.CouplingFlow(transform="spline")
```

```
workflow_spline = bf.BasicWorkflow(  
    inference_network=inference_network_spline,  
    summary_network=summary_network,  
    adapter=adapter_std,  
    simulator=simulator)
```

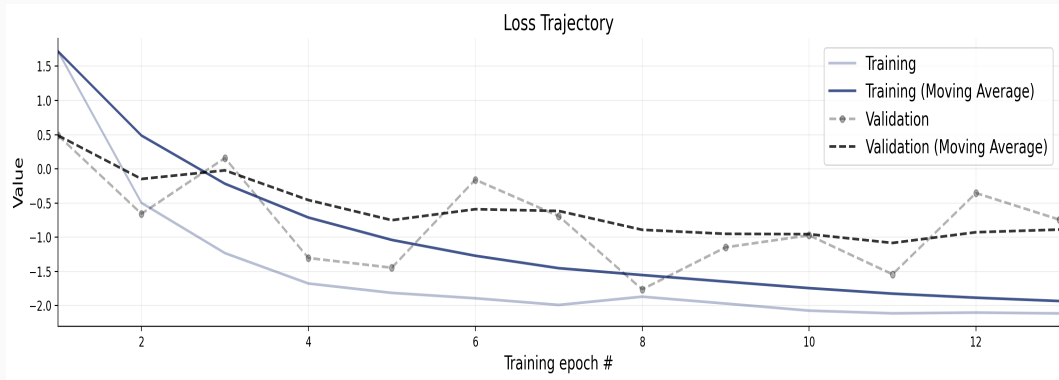
# Evaluation

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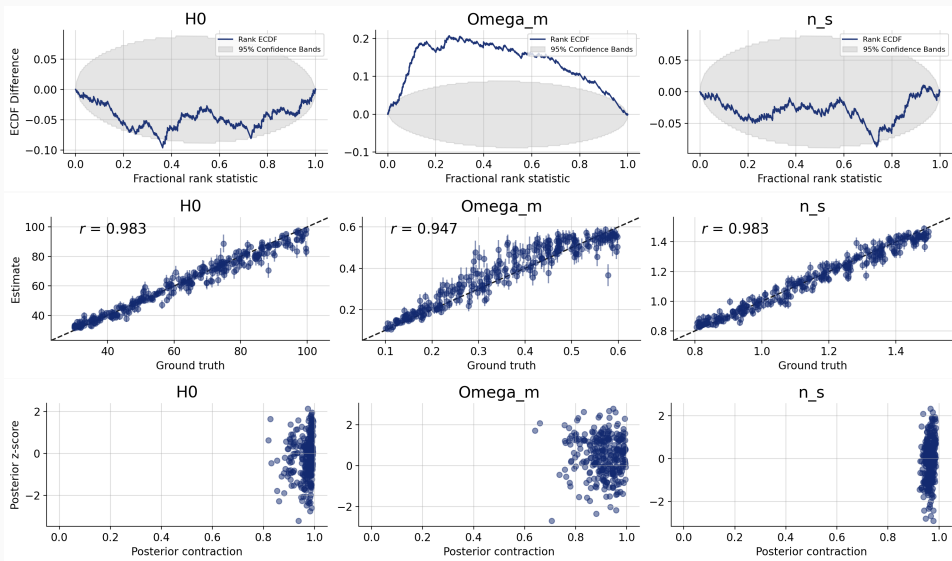
- Training data:  $10^5$  samples
- Epochs: 50, 200 batches per epoch, with early stopping
- Batch size = 128
- Mode: offline training
- Time: 6+ hours per model



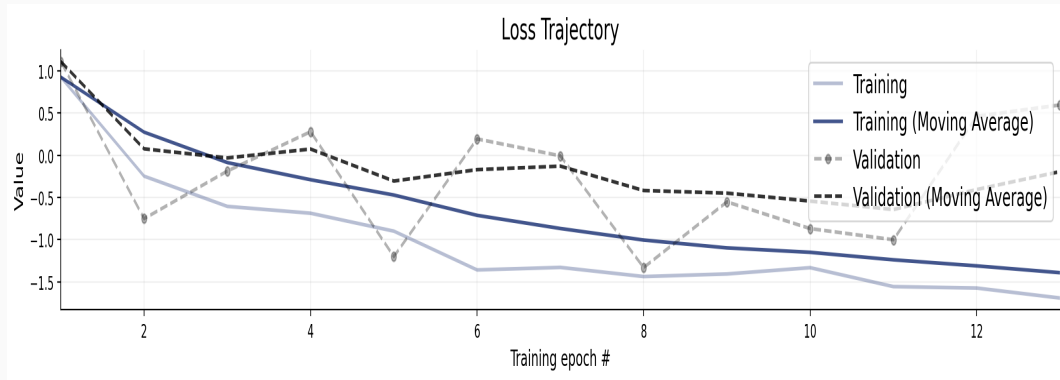
# Model diagnostics - Affine coupling



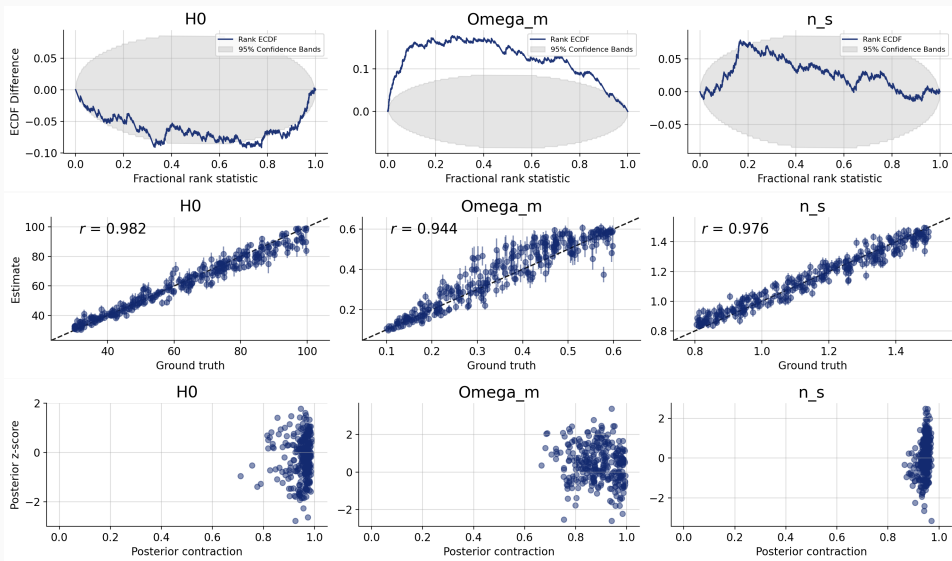
# Model diagnostics - Affine coupling



# Model diagnostics - Spline coupling



# Model diagnostics - Spline coupling



## Summary

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# Discussion

- Both models converged successfully. Results from both models are quite similar;
- For both the models, ECDF difference  $H_0$  lies within 95%CI; well-calibrated, however it is slightly underestimating for spline model. For both models,  $\Omega_m$  is overestimated.
- For affine coupling model,  $n_s$  is within 95%CI, whereas in case of spline coupling, it is slightly overestimating.
- Recovery scatter plots, for both models, show tight alignment along the diagonal. Posterior means closely match true values for all three parameters.
- The contraction plots are mostly centered around zero, indicating balanced uncertainty.
- Early stopping is worth it to save computation time.
- For untransformed data the recovery plots were showing almost constant posterior estimation for all values of each parameter.

- Take smaller range for prior values;
- Standardize the inputs for training on untransformed value;
- Other priors: e.g. normal distribution, beta-distribution;
- Use different inference network architecture, e.g. Flow Matching()